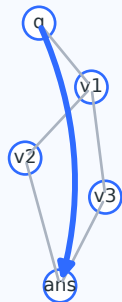


Graph-based methods as chain-structure estimators

Graphs improve observability through expansion and fusion through path constraints.

chain prior



path/subgraph
can be audited

Question-specific heterogeneous graphs

Build and expand query-conditioned evidence subgraphs

• GRAFT-Net

• PullNet

• DFGN / HGN

• SAE / HDE

LM-GNN coupled reasoners

Couple language representations with graph message passing

• QA-GNN

• GreaseLM

• UniKGQA

• ReaRev / NSM

LLM-built and LLM-traversed graphs

Construct, retrieve, and traverse graph memories with LLM-guided operations

• ToG / ToG 2.0

• RoG

• GraphRAG

• HippoRAG / LightRAG

Takeaway: graph quality is an upstream bottleneck - missing nodes or noisy links cap every downstream gain.